Question: What colour change is observed for a positive Bradford test?

Answer: The protein solution changes from brown to blue in colour.

Part 1

Aim: To estimate the concentration of proteins using Bradford Reagent by observing the colour change. (Lab Protocol)

Procedure:

1. Weigh a known amount of standard protein sample using weighing machine.
2. Dissolve it in water to prepare a stock solution. (e.g. : 1 mg/ml)
3. Using pipette, prepare a series of dilutions (e.g.: 0, 20, 40, 60, 80, 100 mug/ml) in test tubes.
4. Adding Bradford Reagent (typically, 1mL) to each of the test tubes.
5. Let the tubes stand at room temperature for about 10 minutes.
6. Observe the colour change. The solutions will turn from brown to blue, depending on protein concentration.
7. Use our eyes to note the intensity of the blue colour.

Part 2

Aim: To estimate the concentration of proteins using Bradford Reagent by observing the colour change.

Set-up: Bradford Reagent, Water and Protein Standard are pre-filled in dedicated rectangular holding chambers. These chambers have motorized pumps to dispense precise volumes. Syringe mechanism connected to a servo pump can extract specific reagent amounts and inject into test tubes.

Procedure:

1. The auger drills into the soil to collect subsurface sample. It is operated using a motor.
2. The scoop attached to the robotic arm (with spoon) collects the loose sample.
3. The robotic arm drops the sample into a circular holding chamber for initial storage.
4. A motor + ratchet mechanism or tilts the chamber to pour a portion into a test tube.
5. Syringe + Pump mechanism is used to add a fixed amount of water from its chamber into this test tube.
6. Syringe + Pump mechanism is used to add a fixed amount of Bradford reagent from its chamber into this test tube.
7. Mixing occurs via controlled vibration using motors.
8. The solution is then transferred into cuvette for optical reading.
9. A camera is positioned to observe the cuvette. It captures the colour change from brown to blue, that indicates the presence of protein molecules.